

A photograph of a SpaceX Falcon Heavy rocket launching from the Kennedy Space Center. The rocket is ascending vertically, leaving a large, billowing plume of white smoke and fire. The launch pad structure is visible on the right side of the frame. The sky is a clear, deep blue.

From Insights to Prediction: Data Analysis and Machine Learning on SpaceX Landings

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Executive summary

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- Interactive dashboard with Plotly Dash
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- EDA results
- Interactive Analytics
- Predictive Analysis Results

Introduction

Project background & context

- The commercial space age is driven by affordability and efficiency.
- SpaceX Falcon 9 pioneered reusability, cutting launch costs to \$62M vs. \$165M+ for competitors.
- Successful first-stage recovery is critical for lowering costs and sustaining innovation.
- Predicting reusability helps in:
 - Improving launch cost estimations
 - Optimizing mission planning
 - Supporting strategic business decisions
- This project applies machine learning to forecast first-stage landing success, aligning with SpaceX's vision of affordable space exploration.

Problem statement

- Has SpaceX's landing success rate improved over time, reflecting increased operational proficiency?
- How do factors such as payload mass, launch site, flight history, and orbit influence landing success?
- What is the performance of machine learning models on unseen data, and which classification model is most suitable for potential deployment?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Combined the web scrapping data: Rocket Launch Data from SpaceX public API and SpaceX Wikipedia page
- Perform data wrangling
 - Identified orbit frequencies, successful vs failed landings, and labelled missions accordingly
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Built several classification models, tuned using GridSearchCV, and evaluated the models using several metrics

Data Collection

The data was collected through a combination of the **SpaceX REST API** and **web scraping** from SpaceX's Wikipedia page. This approach ensured access to both comprehensive historical records and the most up-to-date launch information. The features obtained from each method are listed below.

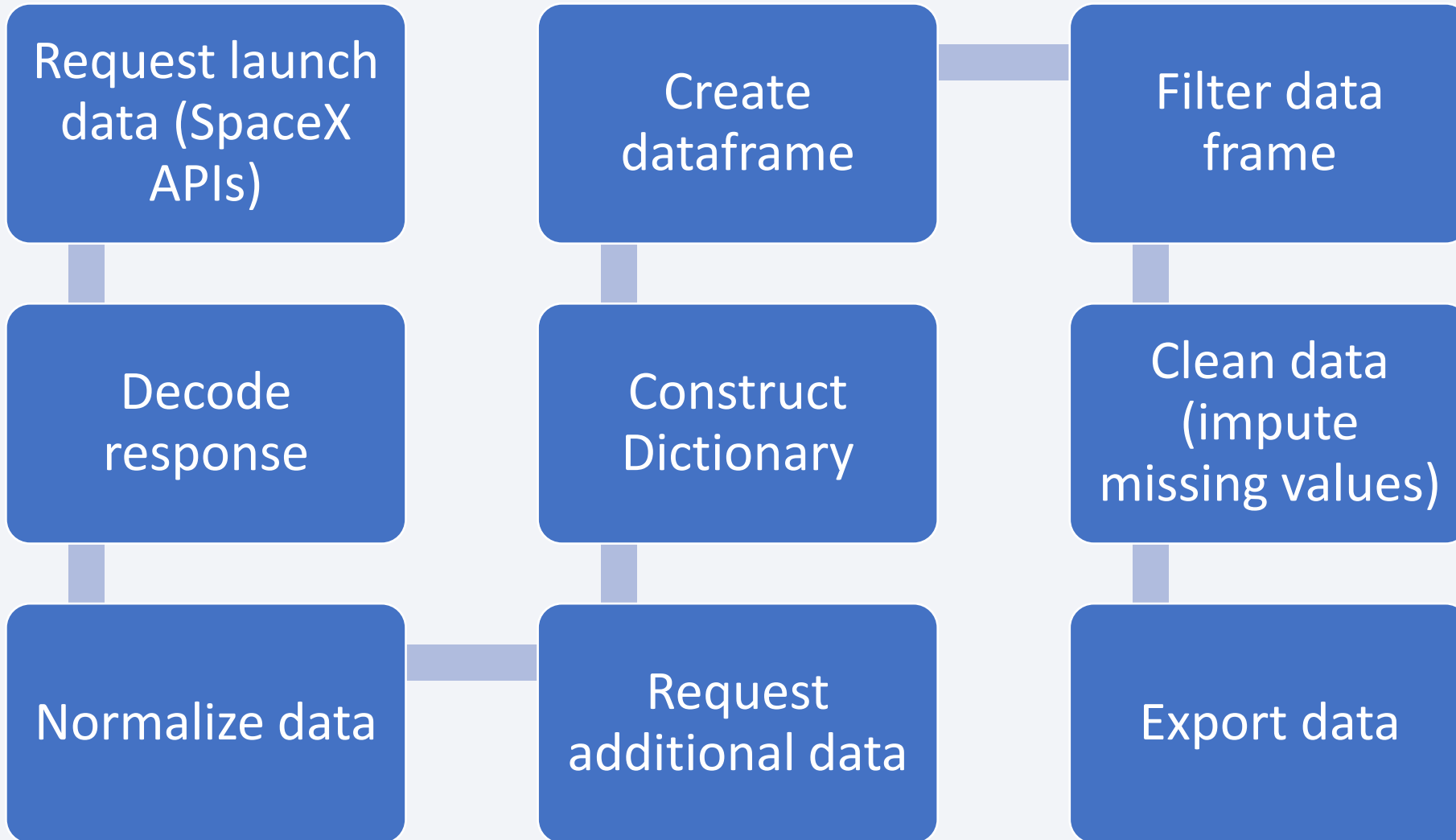
SpaceX REST API

FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude

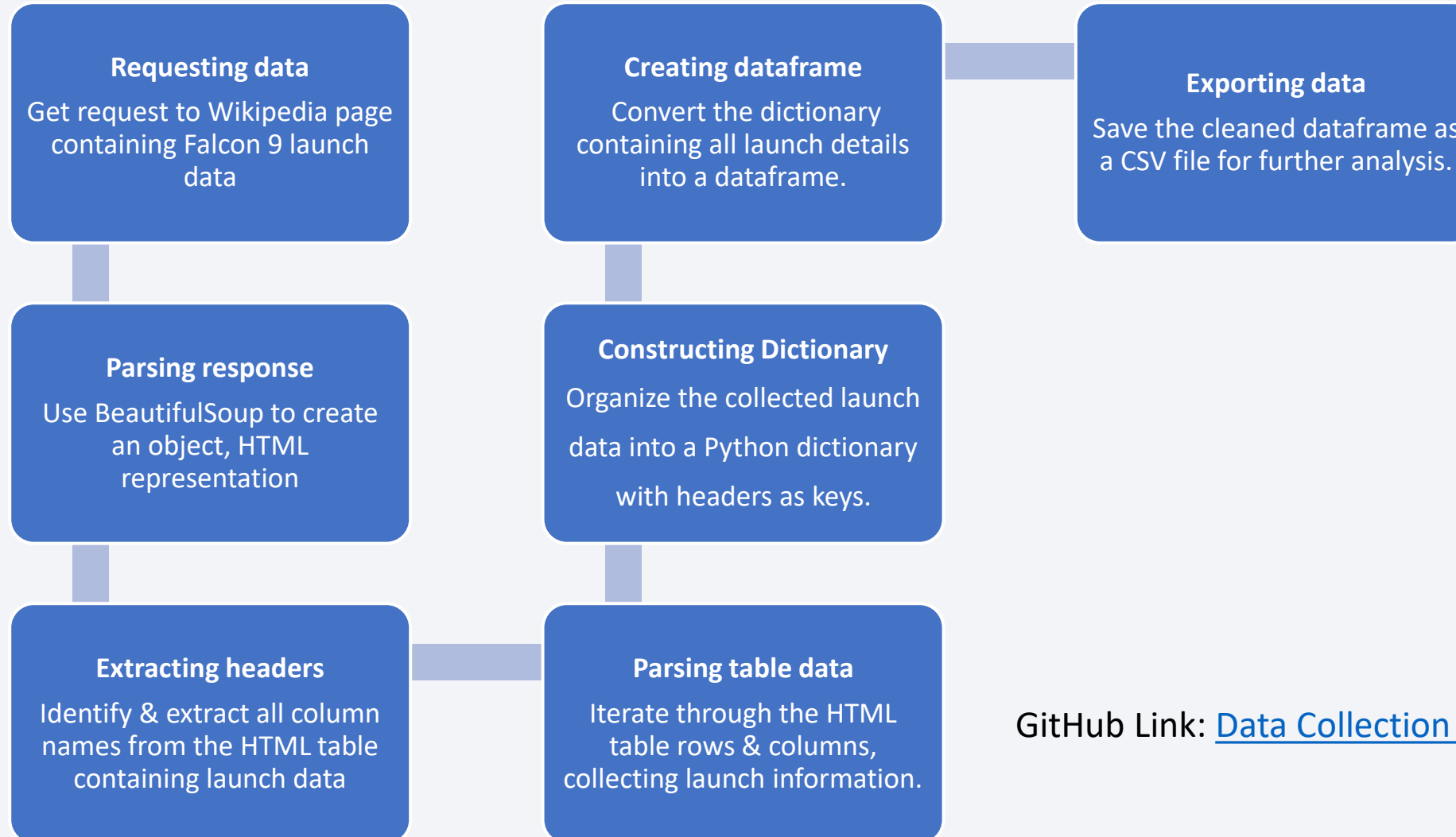
Web Scraping

Flight No., Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, Time

Data Collection – SpaceX API

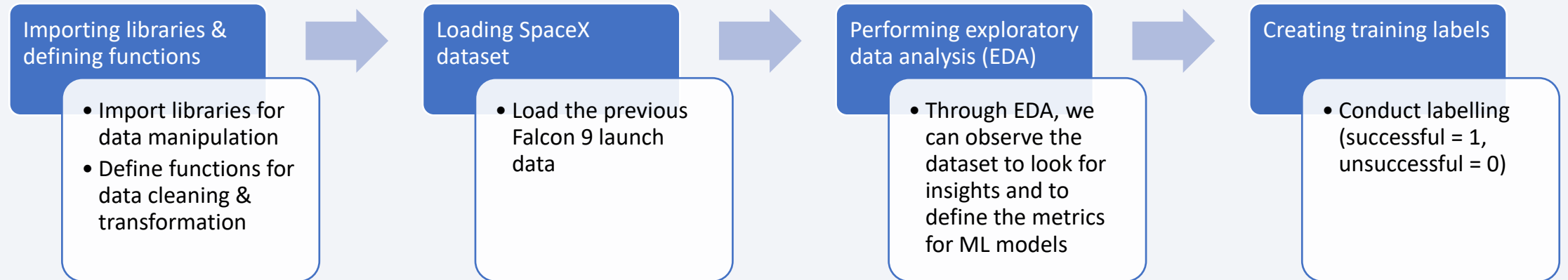


Data Collection - Scrapping



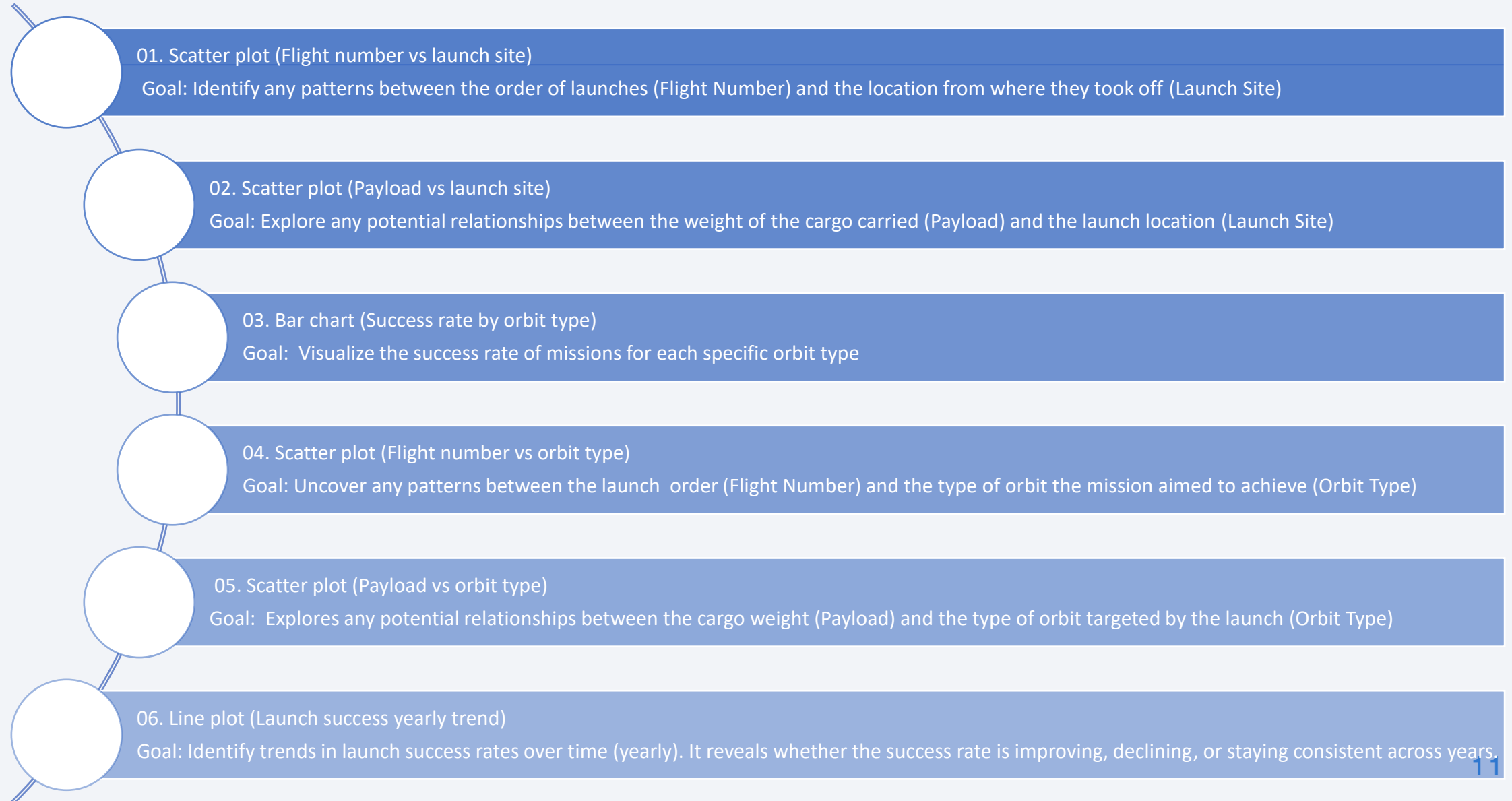
GitHub Link: [Data Collection - web scrapping](#)

Data Wrangling



GitHub Link: [Data Wrangling](#)

EDA with Data Visualization

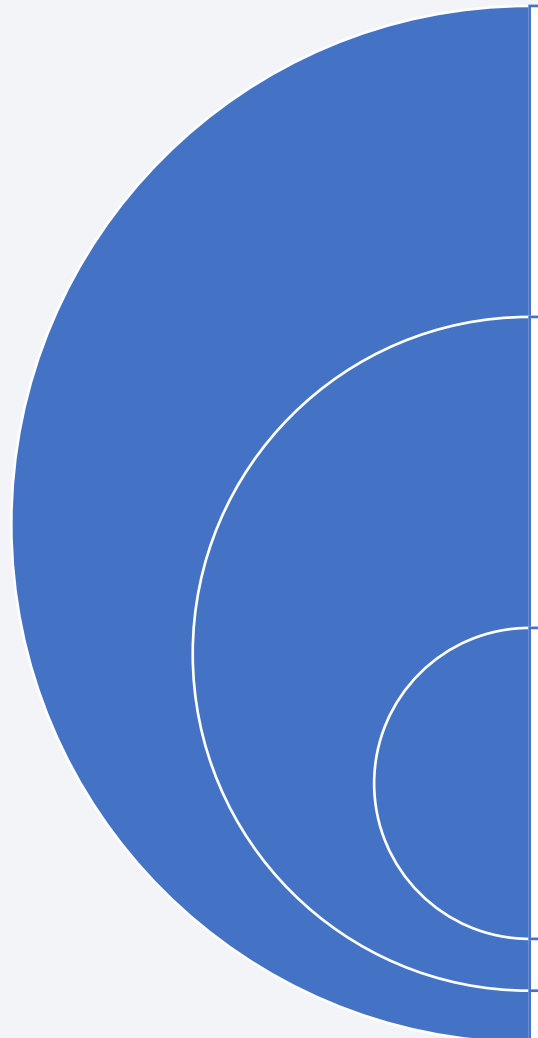


EDA with SQL

- Loaded data set into IBM DB2 Database.
- Queried using SQL Python integration.
- Queries were made to get a better understanding of the dataset.
- Queried information about launch site names, mission outcomes, various pay load sizes of customers and booster versions, and landing outcomes

GitHub Link: [EDA with SQL](#)

Build an Interactive Map with Folium



Launch sites	• Markers with circles highlight all launch site locations. Text labels provide additional information. These markers can also showcase the geographical distribution of launch sites and their proximity to the equator and coastlines.
Launch Outcomes	• Green and red markers represent successful and failed launches, respectively. Marker clusters group these markers, allowing viewers to identify launch sites with high success rates at a glance.
Distances	• Lines connect a specific launch site (e.g., KSC LC-39A) to nearby features like railways, highways, coastlines, and the closest city. These lines illustrate the proximity of launch sites to essential infrastructure and population centres.

Build a Dashboard with Plotly Dash

Dropdown list

- This list allows users to filter data by a specific launch site. By selecting a site, the dashboard focuses on that location's launch information.

Pie Chart

- The pie chart displays the total number of successful launches across all sites. When a specific site is chosen from the dropdown, the pie chart dynamically updates to show the success and failure rates for that particular location.

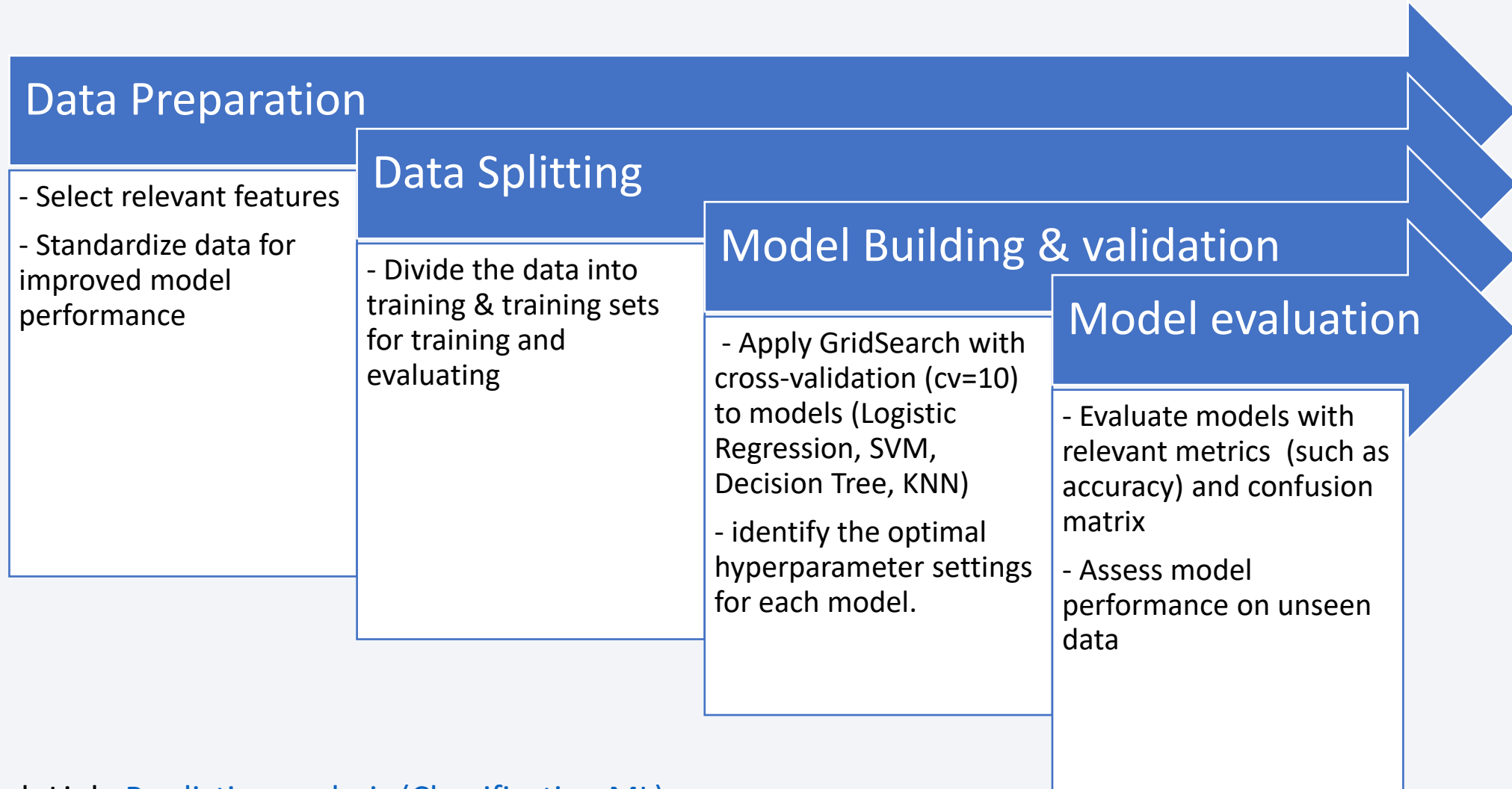
Slider

- This slider enables users to select a specific range of payload mass. This allows for focused analysis on how launch success rates correlate with the weight of the cargo carried

Scatter plot

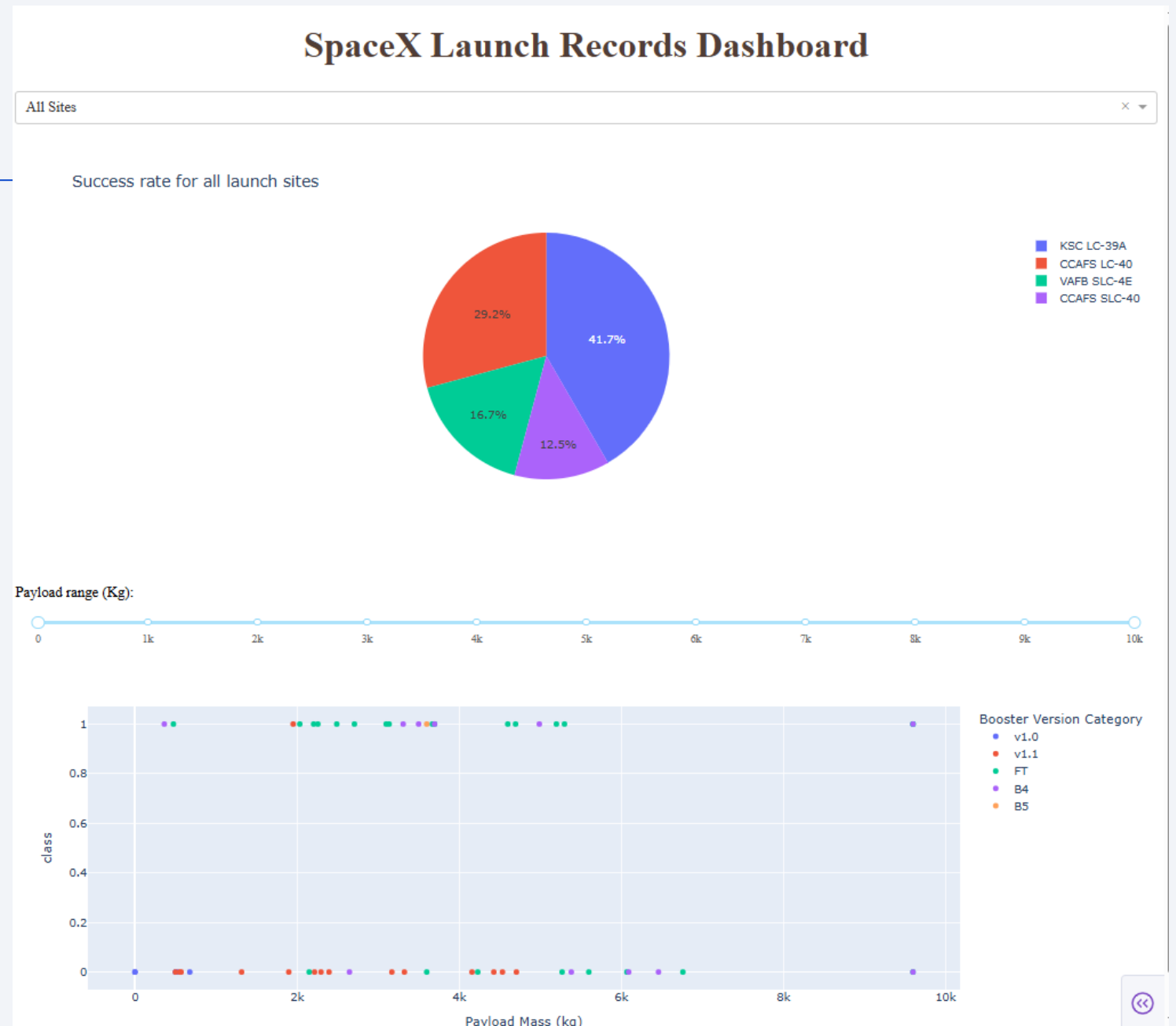
- This scatter plot visualizes the relationship between payload mass (weight) and launch success rate for different booster versions. By filtering the data with the slider and potentially the dropdown, users can explore these correlations for specific launch sites or booster types.

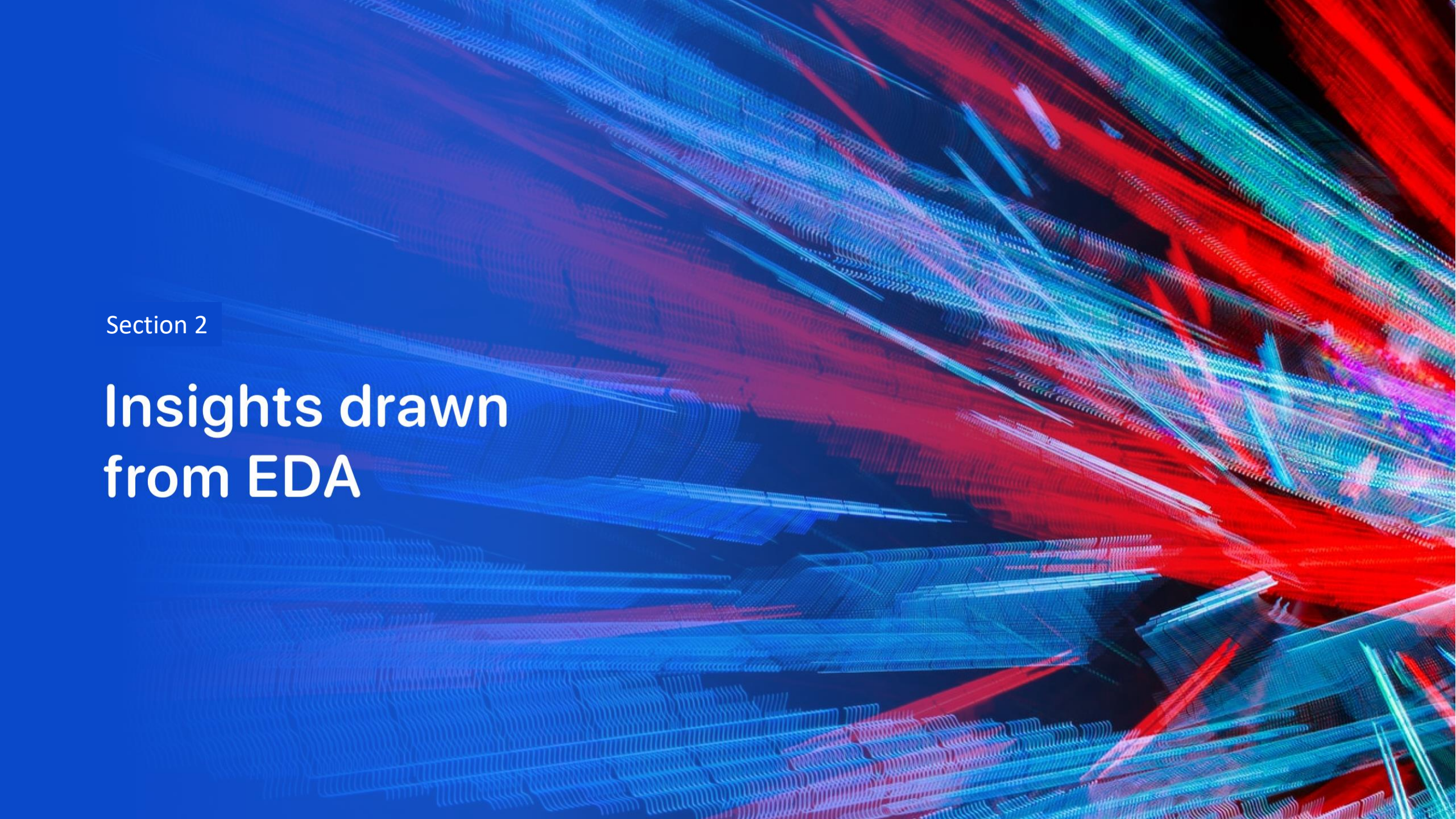
Predictive Analysis (Classification)



Results

This is a preview of the Plotly dashboard. The following slides will show the results of EDA, Interactive analytics demo (dashboard), and finally the results of our model with about 83% accuracy.



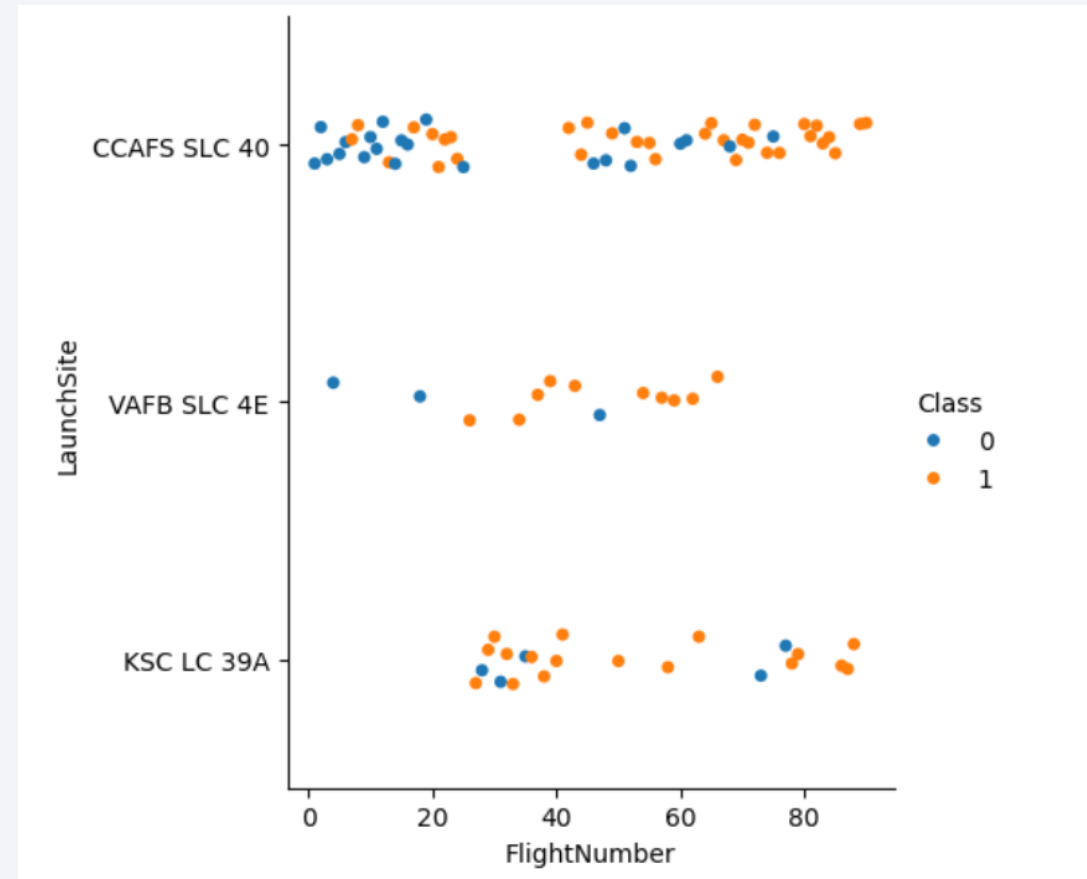
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

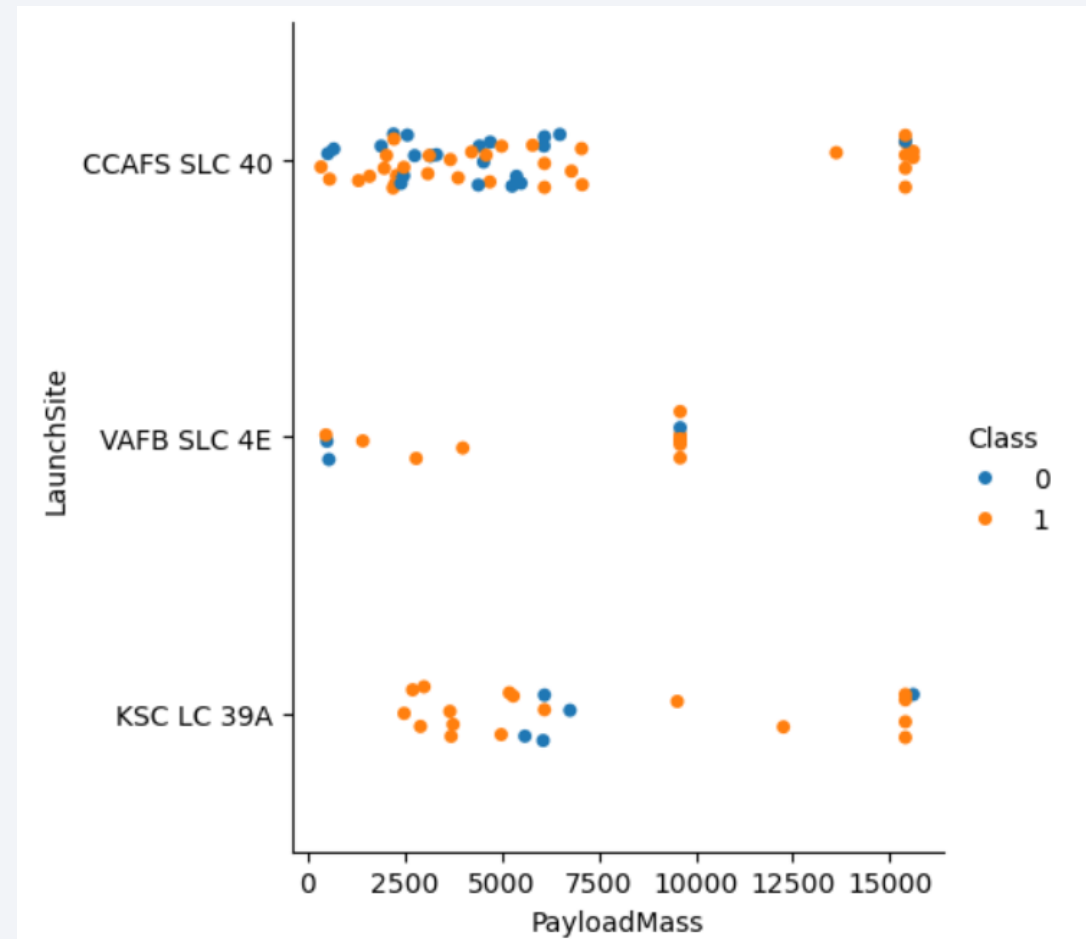
- The graph suggests an increase in success rate over time (indicated in Flight Number). Previously they have many unsuccessful results.
- CCAFS launches most, but VAFB & KSC have higher success.
- This suggests SpaceX is learning and improving over time.



Yellow: Successful launch
Blue: Unsuccessful launch

Payload vs. Launch Site

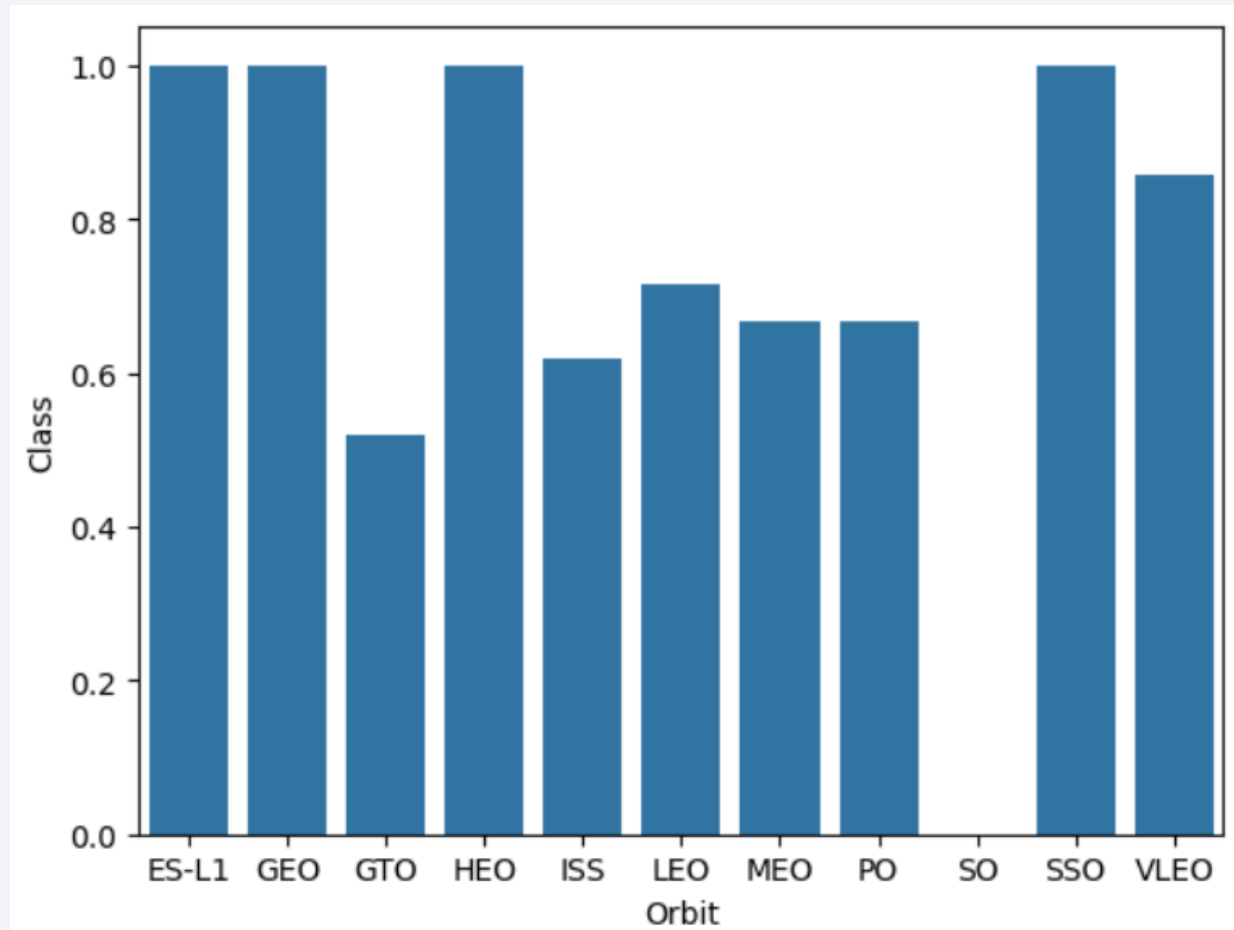
- Heavier payloads have higher tendency to see higher success across sites, with KSC excelling for lighter payload ones.



Yellow: Successful launch
Blue: Unsuccessful launch

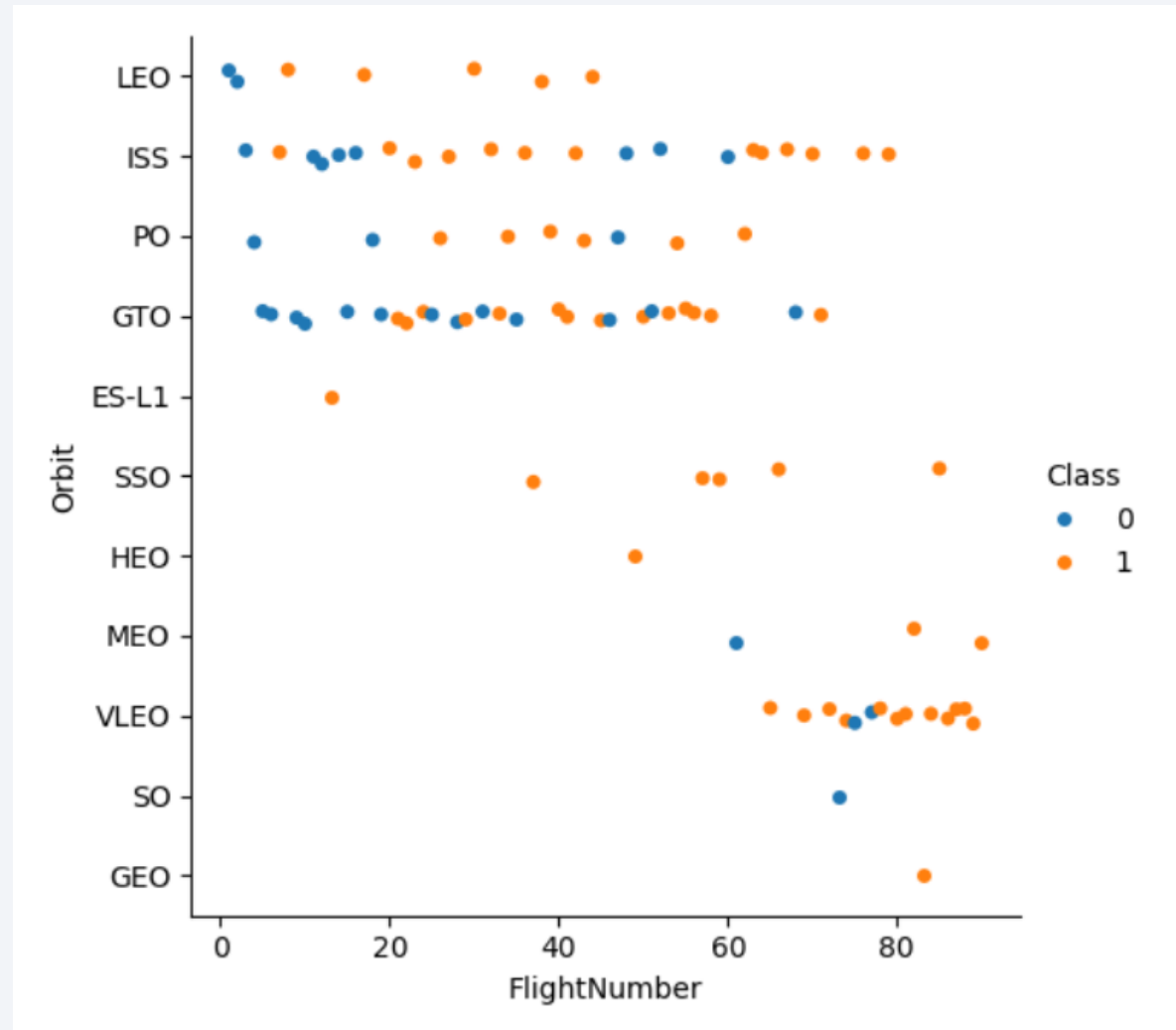
Success Rate vs. Orbit Type

- According to the bar chart the following Orbits show a 100% success rate which include ES-L1, GEO, HEO and SSO.
- Where orbits GTO, ISS, LEO, MEO, PO and VLEO fall under 50-80% success rate.
- Oppositely, SO showed 0% success rate.



Flight Number vs. Orbit Type

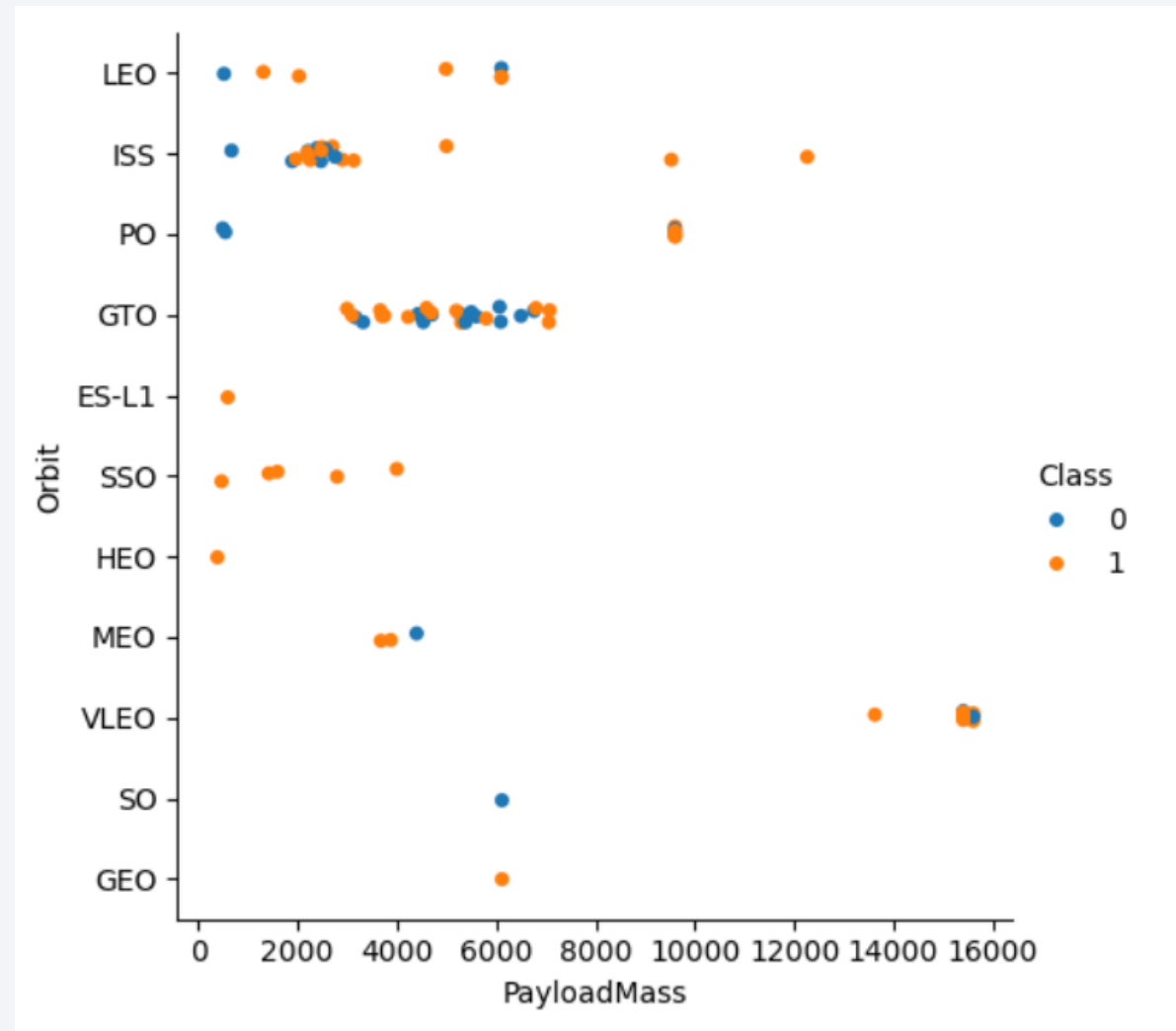
- Launch Orbit preferences changed over Flight Number.
- The LEO orbit the Success appears related to the number of flights (its performance gets better).
- Beside that, there seems to be no relationship between flight number when in GTO orbit.



Yellow: Successful launch, Blue: Unsuccessful launch

Payload vs. Orbit Type

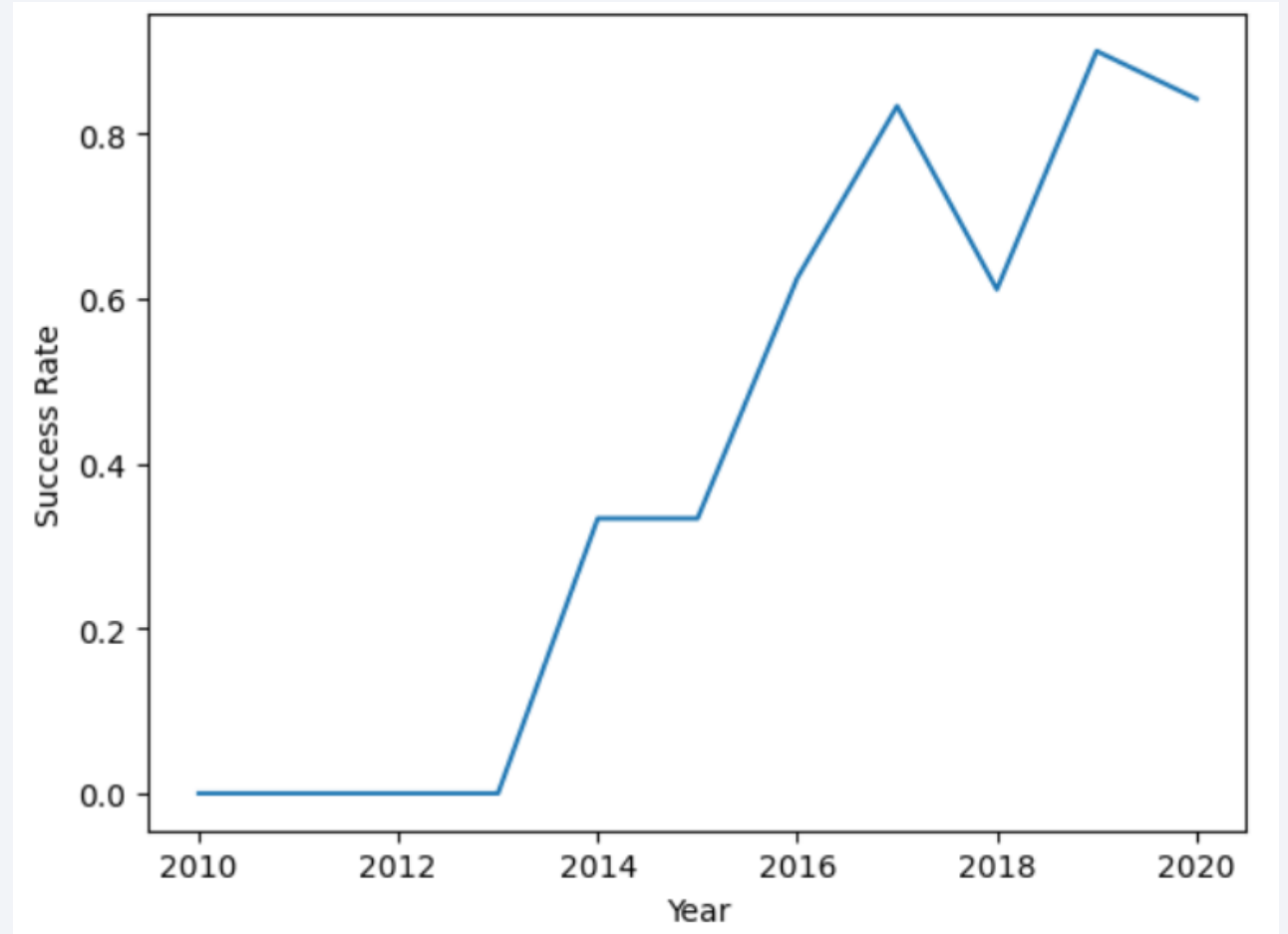
- Payload mass seems to correlate with orbit. Except GTO, as both successful and unsuccessful launch are both present in relatively proportional number.
- In addition, LEO and SSO seem to have relatively low payload mass.



Yellow: Successful launch, Blue: Unsuccessful launch

Launch Success Yearly Trend

- Success rate generally increases overtime from 2013 to 2020, with a slight dip in 2018.
- In addition, success in the most recent years about 80%.



All Launch Site Names

```
%%sql
select distinct Launch_Site from SPACEXTBL
```

* sqlite:///my_data1.db
Done.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- The query aims to display all unique launch sites used which are

1. CCSFS LC-40
2. VAFB SLC-4E
3. KSC LC-39A
4. CCAFS SLC-40.

Launch Site Names Begin with 'CCA'

```
%%sql
select * from SPACEXTBL
where Launch_Site like 'CCA%'
limit 5
```

* sqlite:///my_data1.db
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The following query displays the 5 records of launches done from all sites which name begin with 'CCA'

Total Payload Mass

```
%%sql
select SUM(PAYLOAD_MASS__KG_) as total_payload_mass from SPACEXTBL
where "Customer" = "NASA (CRS)";
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

total_payload_mass

45596

- This query sums the total payload mass carried by the NASA (CRS) which is 45596 kg.
- CRS stands for Commercial Resupply Services which indicates that these payloads were sent to the International Space Station (ISS).

Average Payload Mass by F9 v1.1

```
%%sql
select AVG(PAYLOAD_MASS__KG_) as average_payload_mass from SPACEXTBL
where Booster_Version = 'F9 v1.1'
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

average_payload_mass

2928.4

- This query calculates the average payload mass carried by the booster version F9 v1.1 which is 2928.4 kg.
- Average payload mass of F9 1.1 is on the low end of our payload mass range

First Successful Ground Landing Date

```
%%sql  
select min(SPACEXTBL.Date) from SPACEXTBL  
where Landing_Outcome="Success (ground pad)"
```

```
* sqlite:///my\_data1.db  
Done.
```

min(SPACEXTBL.Date)
2015-12-22

This following query returns the first successful ground pad landing date (22nd December 2015)

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%%sql
select Booster_Version from SPACEXTBL
where Landing_Outcome="Success (drone ship)" and PAYLOAD_MASS__KG_ between 4000 and 6000
```

* [sqlite:///my_data1.db](#)

Done.

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

The query returns the four booster versions that had successful drone ship landings with payload mass between 4000 and 6000 kg

Total Number of Successful and Failure Mission Outcomes

```
%%sql
select Mission_Outcome, COUNT(*) from SPACEXTBL
group by 1
```

* [sqlite:///my_data1.db](#)

Done.

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- The following query groups and displays total number of mission outcomes whether it success or failure.
- Overall, SpaceX appears to achieve its mission outcome nearly about 98% of the time.
- Unfortunately, one launch has been failed in flight.
- Interestingly, one launch has unclear result.

Boosters Carried Maximum Payload

```
%%sql
select Booster_Version from SPACEXTBL
where PAYLOAD_MASS__KG_ = (select max(PAYLOAD_MASS__KG_) from SPACEXTBL)
```

* sqlite:///my_data1.db

Done.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- The following query gives a list of Booster Versions carrying maximum payload.
- These booster versions are very similar, and all are of the F9 B5 B10xx.x variety.
- This perhaps indicates payload mass correlates with the booster version that is used.

2015 Launch Records

```
• %%sql
  select substr(Date, 6, 2) as month, Date, Booster_Version, Launch_Site from SPACEXTBL
  where Landing_Outcome = 'Failure (drone ship)' and substr(Date,0,5)='2015'
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

month	Date	Booster_Version	Launch_Site
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40

- This query above returns the Month, Landing Outcome, Booster Version, Payload Mass (kg), and Launch site of 2015 where the landing outcome was failure on drone ship.
- There were two failed occurrences.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
select Landing_Outcome, COUNT(Landing_Outcome) as count_Outomes from SPACEXTBL
where Landing_Outcome like 'Success%' and SPACEXTBL.Date between '2010-06-04' AND '2017-03-20'
group by 1
order by 2 DESC
```

✓ 0.0s

* [sqlite:///my_data1.db](#)
Done.

Landing_Outcome	count_Outomes
Success (drone ship)	5
Success (ground pad)	3

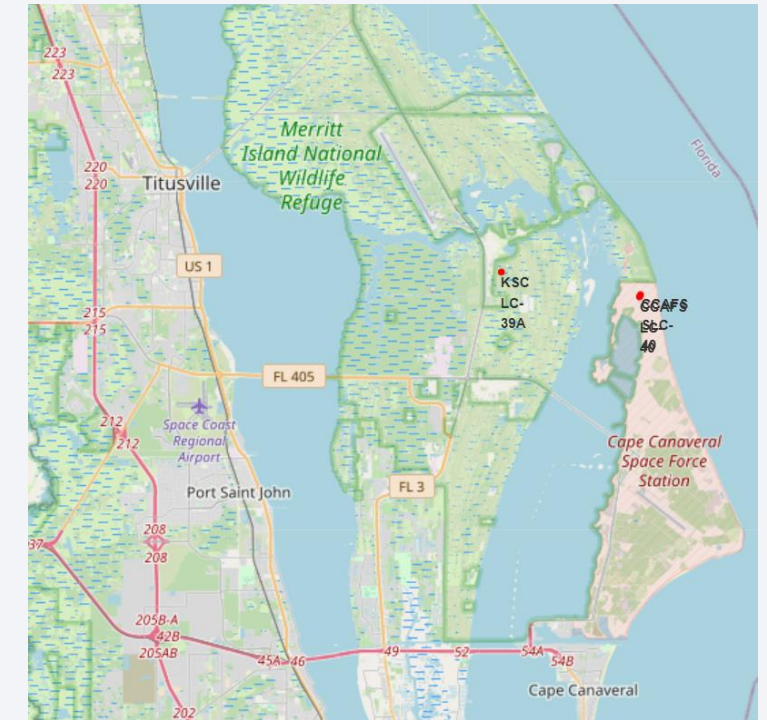
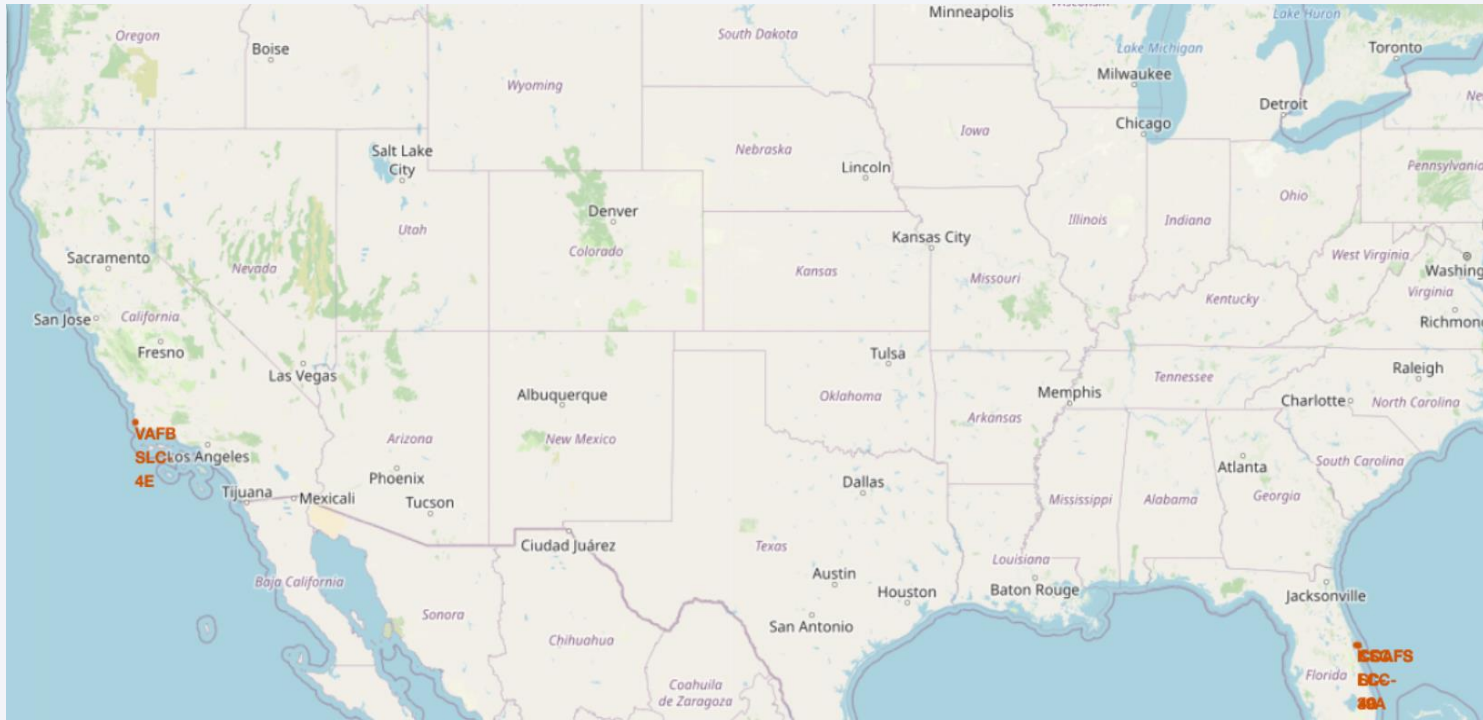
- This following query returns a list of successful landings between specific dates (2010-06-04 and 2017-03-20) and group them by landing outcomes.
- There are two types of successful landing outcomes: drone ship and ground pad landings.
- Overall, there were 8 successful landings in total in this particular period.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

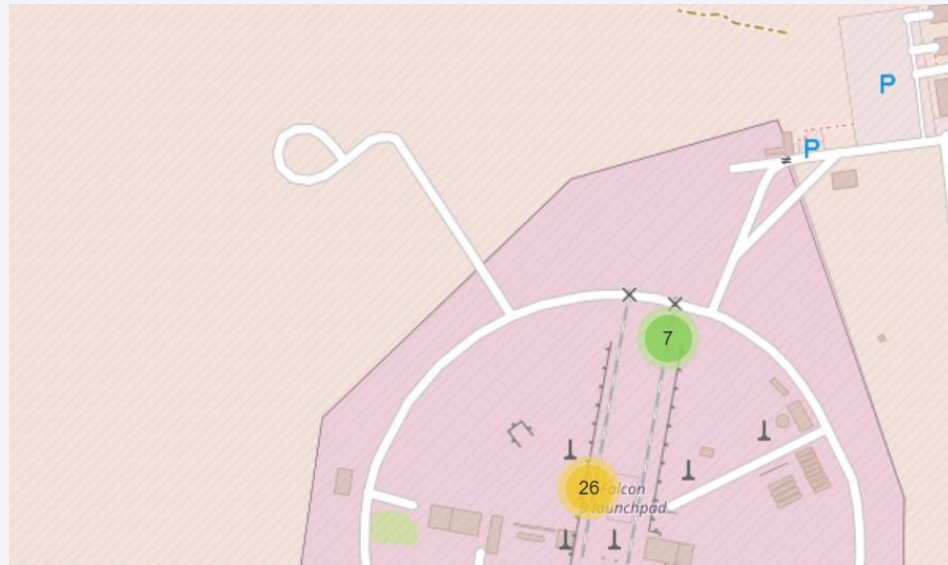
Launch Sites Proximities Analysis

Launch sites location markers on a global map

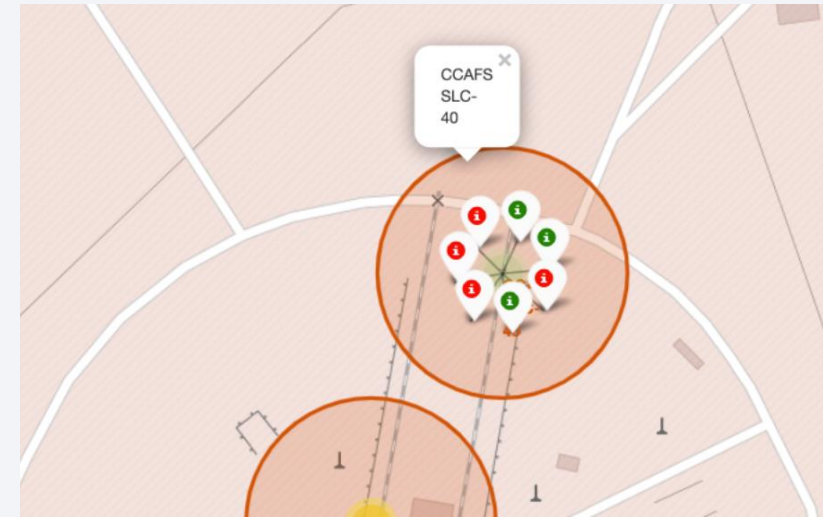


- The left map shows all launch sites (relatively to the US map). All launch sites are located near the ocean for simplifying debris mitigation and avoiding citizens residency.
- The right map shows the two Florida launch sites since they are very close to each other.

Color-coded launch outcomes

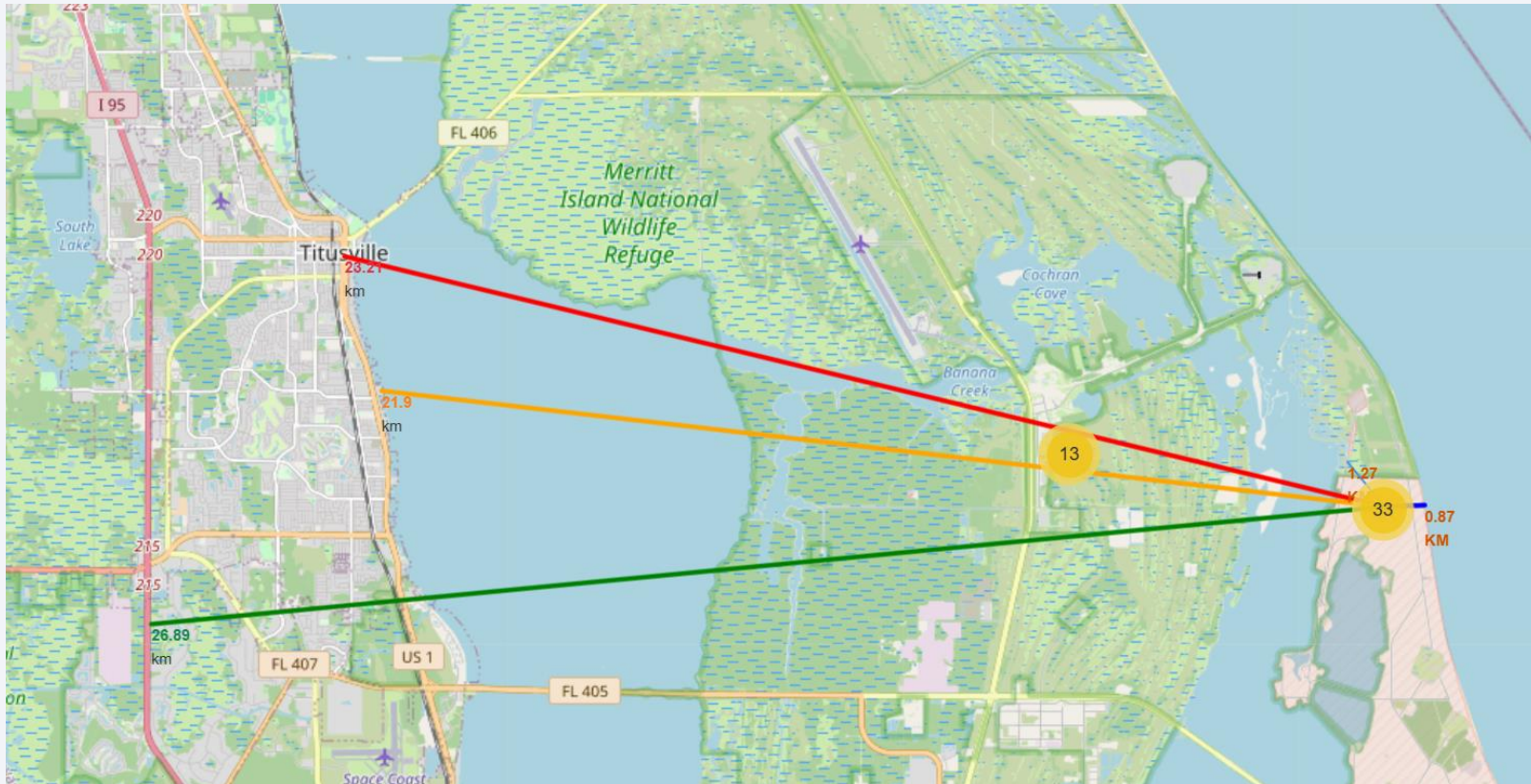


Zoom in

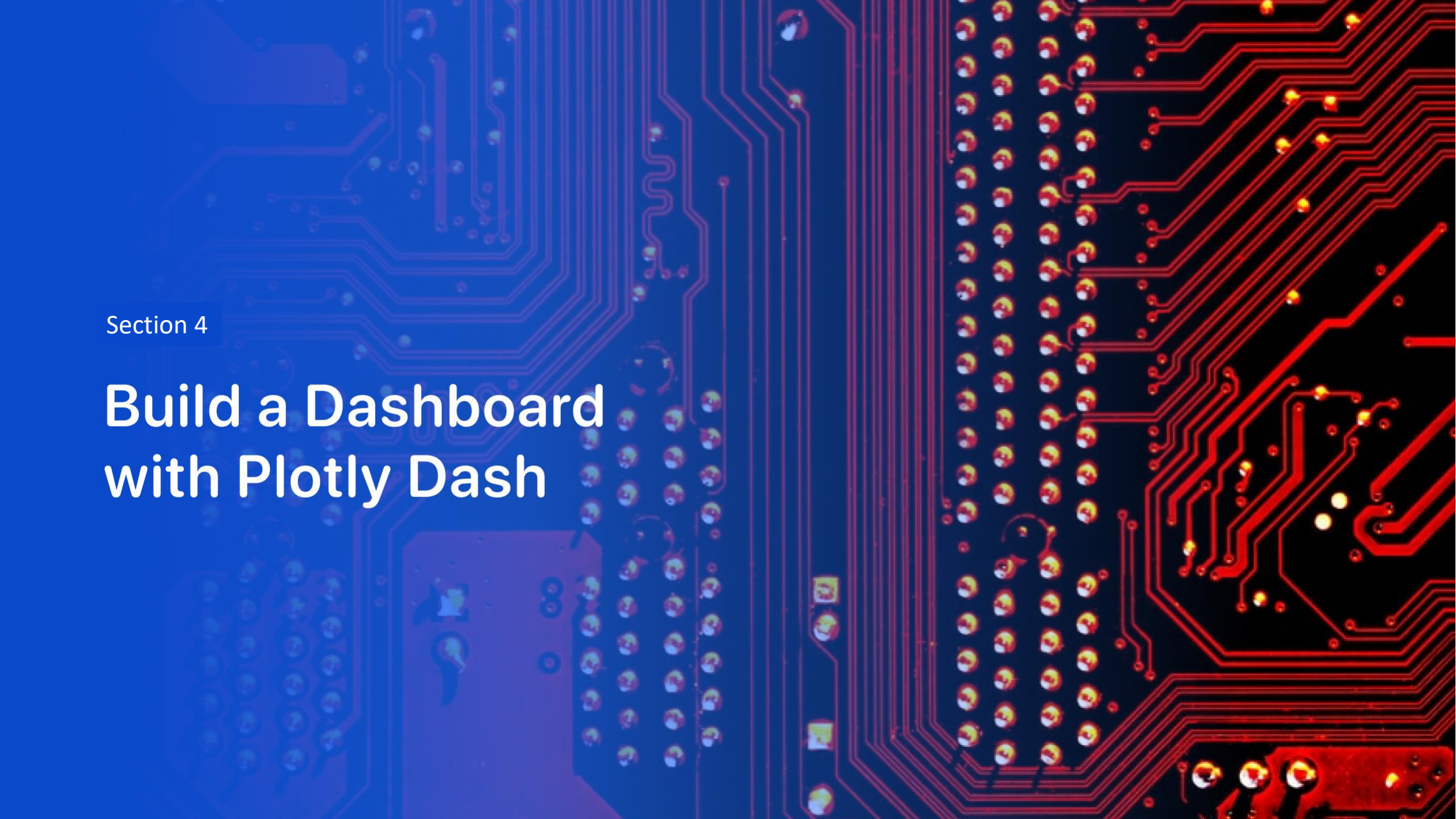


- From the color-labeled markers in marker clusters, we can identify the successful (green icon) and unsuccessful landing (red icon).

Key location proximities from CCAFS LC-40



- The launch site (near marker 33) is near Cape Canaveral, Florida, by the Atlantic Ocean.
- It is only 0.87 km from the coastline, ideal for safety and stage recovery.
- The railway and nearest highway are 21 and 27 km away, ensuring access and safe distance.
- Titusville city lies 23 km away, balancing safety and logistics.



Section 4

Build a Dashboard with Plotly Dash

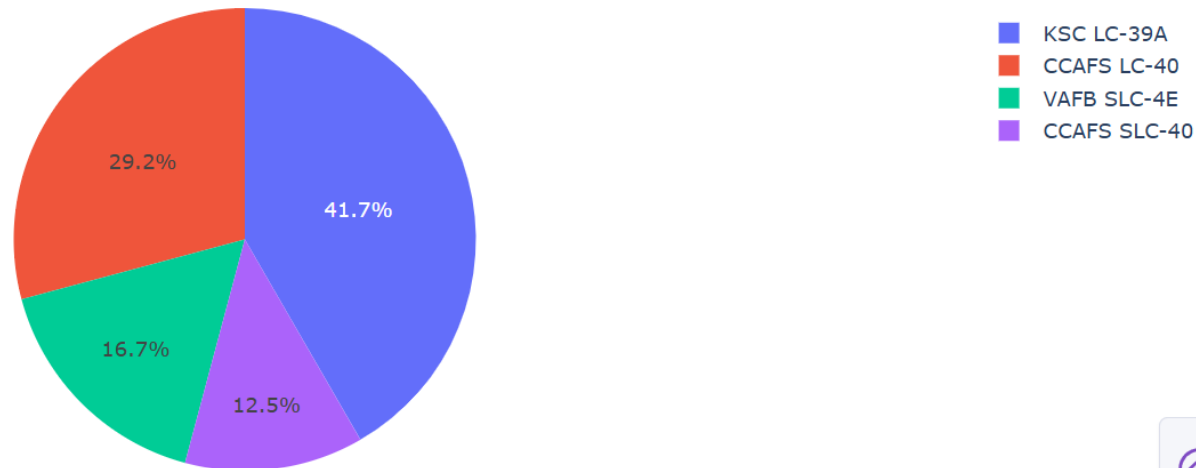
Total success launches by site

SpaceX Launch Records Dashboard

All Sites

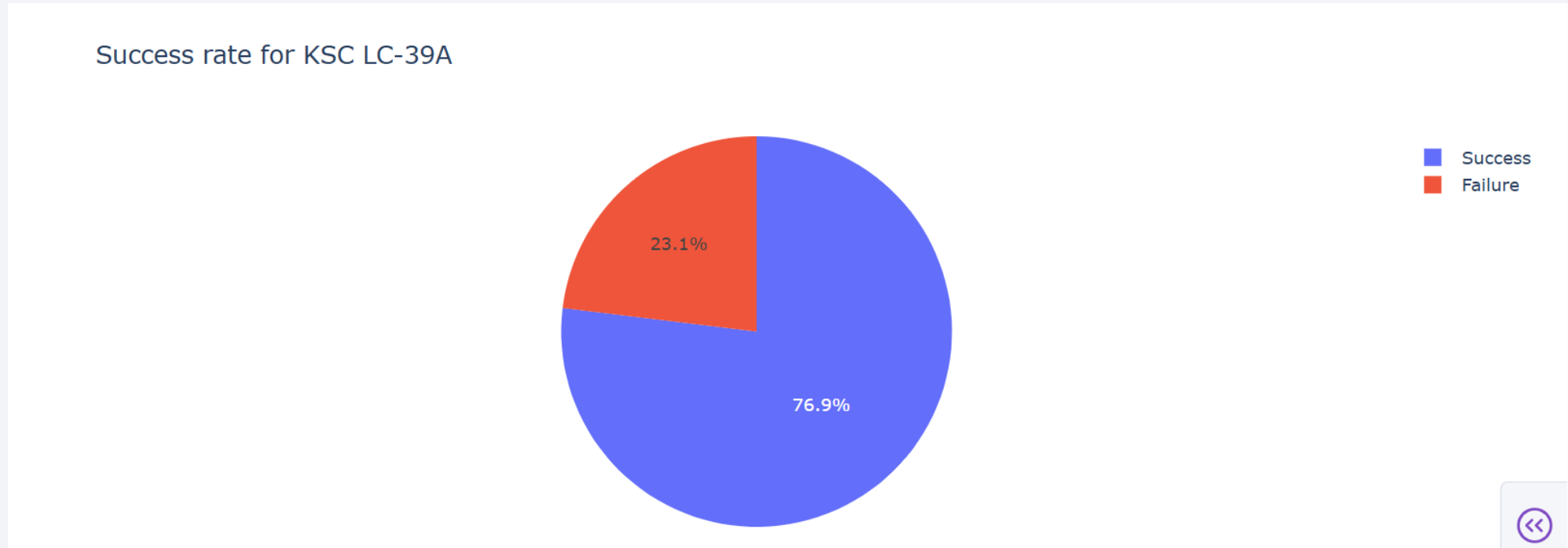


Success rate for all launch sites



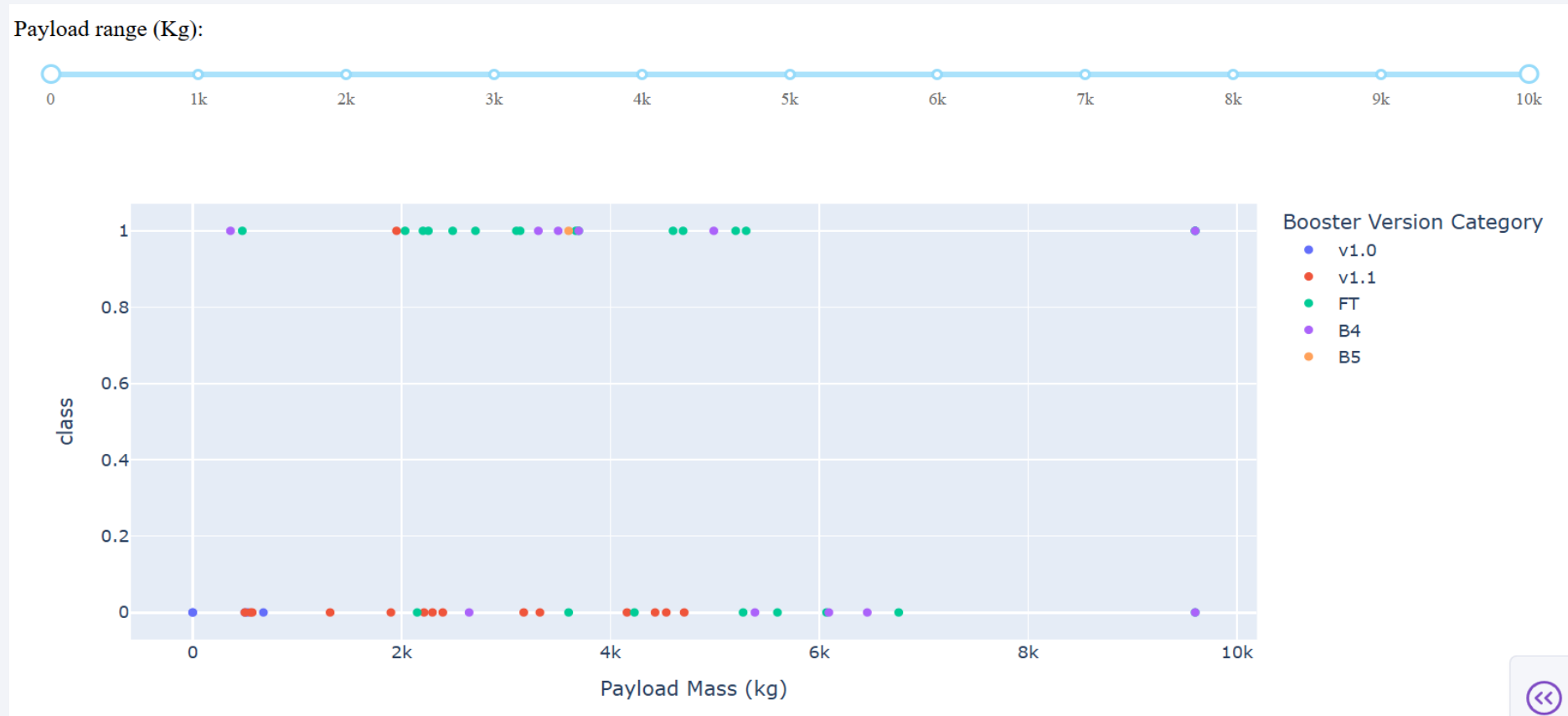
The pie chart displays that KSC LC-39A contributes significantly to the successful launches (about 41.7%) that three times higher than CCAFS SLC-40.

Pie chart for launch site with highest launch success ratio



KSC LC-39A has the highest success rate with launch success ratio is 77 : 23.

Payload vs. Launch outcome for different payloads

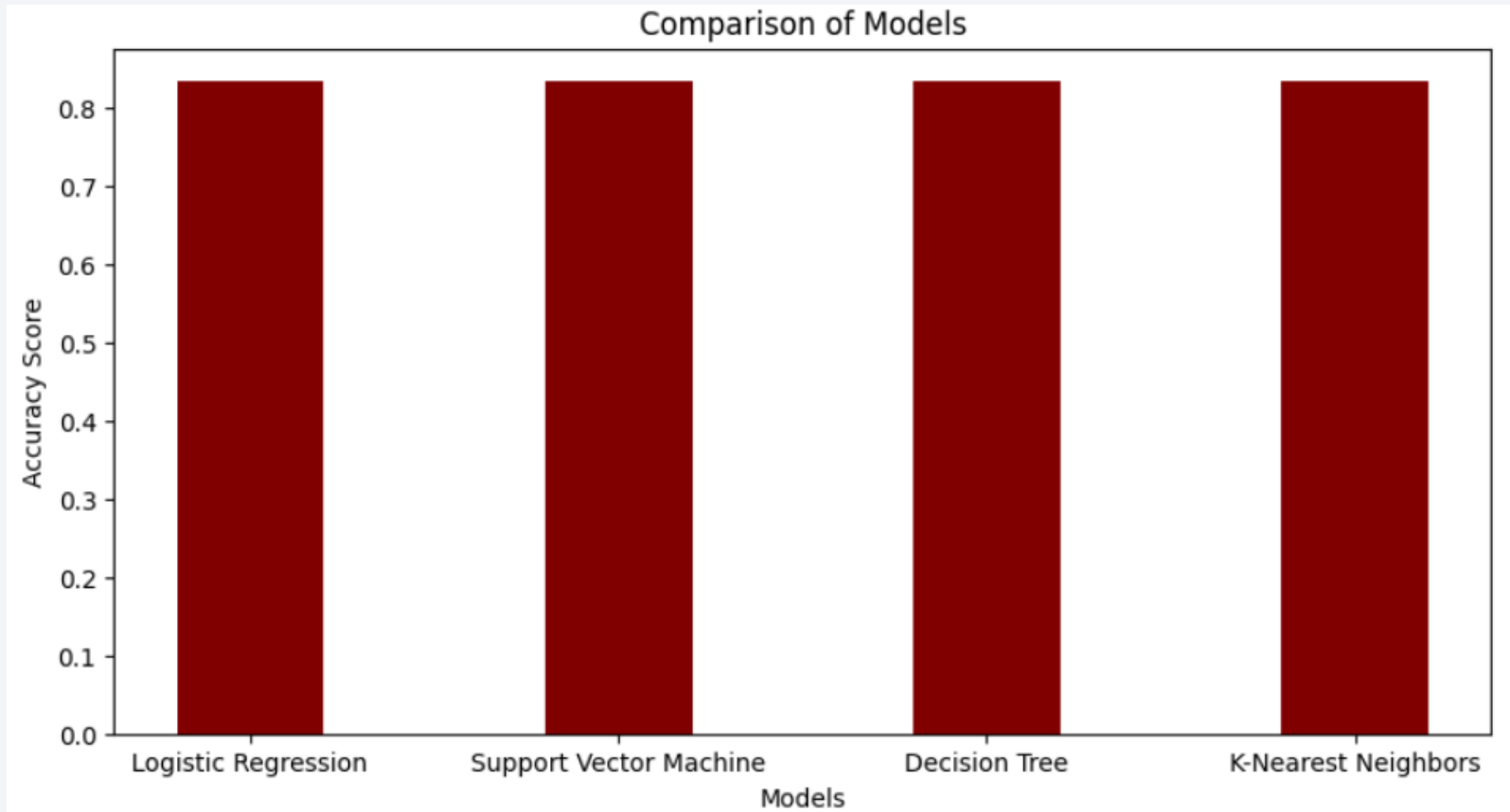


The scatter plot represents launch outcome for payload mass between 0 to 10000 kg for various Booster Version Category

Section 5

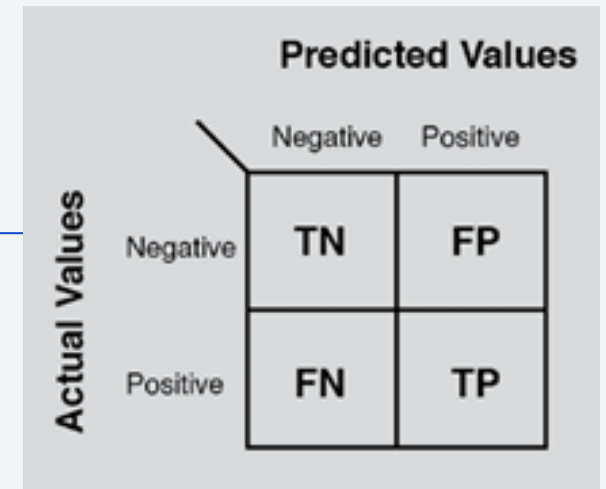
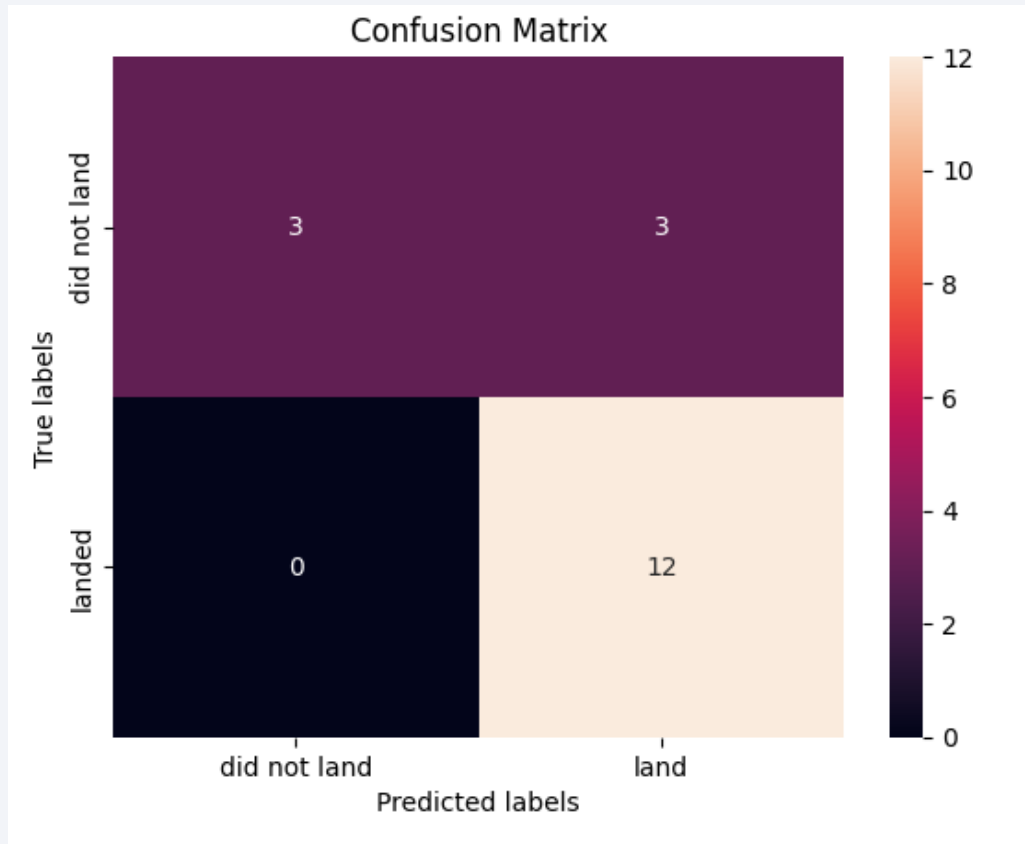
Predictive Analysis (Classification)

Classification Accuracy



- All models had virtually the same accuracy on the test set at 83.33% accuracy.
- It should be noted that test size is small at only sample size of 18.
- We likely need more data to determine the best model.

Confusion Matrix



- Since all models performed the same for the test set, the confusion matrix is the same across all models.
- The models predicted 12 successful landings when the true label were successful landing (True positive)
- The models predicted 3 unsuccessful landings when the true label was unsuccessful landing (True negatives)
- The models predicted 3 successful landings when the true labels were unsuccessful landings (false positives). Our models tend to over predict successful landings.

Conclusions (1)

Landing Success Trend

- Early launches faced challenges, but success rates have significantly improved over time.
- From 2013 to 2020, success rates consistently increased, reaching ~80% in 2020, showing growing SpaceX proficiency.

Factors Affecting Success

- Payload: Heavier payloads generally achieved higher success across sites.
- Launch Site: Kennedy Space Center (KSC) showed better performance for lighter payloads.
- Orbit: Certain orbits (e.g., ES-L1, GEO, HEO, SSO) achieved 100% success, while others (e.g., GTO, ISS, LEO, MEO, PO, VLEO) had success rates between 50–80%.

Conclusions (2)

Machine Learning Model Performance

- All models tested (Logistic Regression, SVM, Decision Tree, KNN) achieved 83.33% accuracy on unseen data.
- Logistic Regression may be preferable due to its interpretability despite similar accuracy across models.
- Further data and study are required to confidently select the best model for deployment.

Appendix

- All external link to Python code snippets, SQL queries, charts, Notebook outputs, or data sets used in the project is provided [here](#).
- Thank you to all the Instructors, Coursera and IBM for providing high-quality lectures.
- Special thanks for the US-ASEAN STIC who sponsored my study for this course.

Thank you!

